JSON Basics

JSON (JavaScript Object Notation) is a lightweight format used to store and exchange data. It is commonly used when sending data between a web browser and a server, and it is widely used in JavaScript, React, APIs, and databases.

Why Use JSON?

- Easy for humans to read and write.
- Easy for computers to parse.
- Supported by almost every programming language.
- Commonly used for APIs and configuration files.

JSON Rules

- Data is written as key-value pairs.
- Keys must be enclosed in double quotes.
- Strings use double quotes.
- Values can be strings, numbers, booleans, null, objects, or arrays.
- JSON does not support comments or functions.

JSON Object Example

```
{
  "name":"John",
  "age":15,
  "isStudent":true
}
```

JSON Array Example

```
[
  {"name":"John","score":90},
  {"name":"Mary","score":95},
  {"name":"Alex","score":80}
]
```

Nested JSON

Objects and arrays can be combined.

```
{

  "name":"Pikachu",

  "types":["Electric"],

  "stats":{"hp":35,"attack":55}

}
```

JSON vs JavaScript Object

JavaScript objects may contain functions, use single or double quotes for strings, and keys do not require quotes. JSON is a data format: keys must use double quotes and functions are not allowed.

JSON.parse()

Convert a JSON string into a JavaScript object.

```
const text='{"name":"John","age":15}';

const student = JSON.parse(text);
```

JSON.stringify()

Convert a JavaScript object into a JSON string.

```
const student={name:"John",age:15};

const text = JSON.stringify(student);
```

When Will You Use JSON?

- Reading data from an API.

- Saving application settings.

- Sending data between frontend and backend.

- Storing sample data for projects.

- Loading product, student, or Pokémon data into a web application.

Summary

JSON is the standard format for exchanging data on the web. Before learning React and databases, understanding JSON objects, arrays, JSON.parse(), and JSON.stringify() is essential.